EE561 Advanced Static Power Conversion

Multilevel Converters — Quiz

15 minutes · closed book · calculator allowed · prepared by Sefer İdacı and M. Aytunç Evlice

Name: Student ID: Date:

Answer all four questions in the space provided; show your reasoning. Each question is worth 25 points (total 100). Questions 1–2 cover Part I (topologies), questions 3–4 cover Part II (modulation and balancing). Useful relations are given at the bottom of page 2.

Question 1 · Device selection

PART I · NUMERICAL DESIGN **25 pts**

A PV string inverter has a DC link of **1100 V**. In steady state a switch may block at most **2/3 of its voltage rating**; the available classes are 650 V, 1200 V, 1700 V and 3300 V.

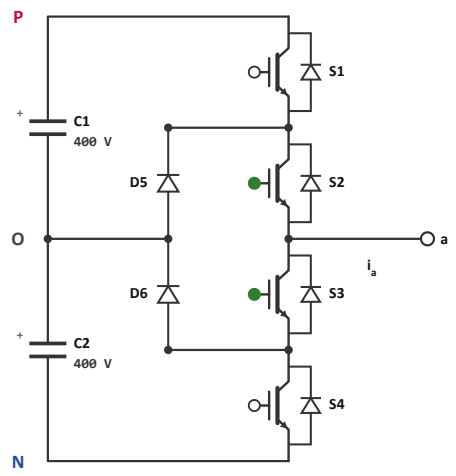
a. Complete the table. [15]

Leg / device	Voltage blocked by the device [V]	Required class [V]
2-level leg, each switch		
3-level NPC leg, each switch		
3-level T-type leg, outer switches S1, S4		
3-level T-type leg, inner switches S2, S3		

b. Give one advantage and one disadvantage of the T-type leg compared with the NPC leg at this DC link. [10]

Question 2 · NPC leg in the zero state

PART I · CONCEPT AND CRITIQUE **25 pts**



3-level NPC leg, V_{dc} = 800 V, state O (gated switches marked green).

The leg is in state **O** and the phase current is negative, $i_a < 0$ (flowing *into* the leg).

a. Draw the current path on the figure and name the conducting devices. [7]

b. Does the leg draw current from the midpoint O or push current into it? [5]

c. Which devices block voltage, and how much? [6]

d. A designer proposes to switch directly from P to N "to save one commutation". Critique this proposal. [7]

Question 3 · DC-bus utilization and min-max injection

PART II · NUMERICAL DESIGN **25 pts**

A three-phase, 3-level NPC converter must supply **380 V line-to-line RMS** at 50 Hz to a three-wire grid from a **600 V** DC link.

a. Calculate the required modulation index $m = V_{1,pk} / (V_{dc}/2)$. [6]

b. Is this possible with sine PWM? With min-max (zero-sequence) injection? Justify. [5]

c. At one instant the normalized references are $r = (0.98, -0.25, -0.73)$. Find the offset r_0 and the modified references r', and show that $r'_a - r'_b = r_a - r_b$. [9]

d. What is the largest line-to-line RMS voltage this converter can produce without overmodulation when min-max injection is used? [5]

Question 4 · Neutral-point balancing

PART II · SIMULATION WHAT-IF AND CRITIQUE **25 pts**

In a 3-level NPC with carrier PWM, a common offset r_0 is added to all three references to balance the split DC link ($\Delta V = v_{C1} - v_{C2}$).

a. Starting from the relation for $\langle i_O \rangle$ below, show how a small r_0 changes the average midpoint current. [8]

b. In the lecture simulation the loop restores ΔV within a few cycles at $\phi = 0°$. Predict what happens at $\phi = 90°$ (pure reactive current) and explain. [8]

c. A team implements $r_0 = +K \cdot \Delta V$, tunes it on an inverter test bench, then ships the converter as an **active rectifier**. Critique this design and propose a fix. [9]

Useful relations

- $m = V_{1,pk} / (V_{dc}/2)$; linear range $m \le 1$ (sine PWM), $m \le 2/\sqrt{3} \approx 1.155$ (min-max injection). $V_{LL,rms} = \sqrt{3} \cdot V_{1,pk} / \sqrt{2}$.
- Min-max offset: $r_0 = -[\max(r_a, r_b, r_c) + \min(r_a, r_b, r_c)] / 2$, $r'_x = r_x + r_0$.
- NPC with carrier PWM: $\langle i_O \rangle = \Sigma_x (1 - |r_x|)\, i_x$ (current drawn from O), $C\, d\Delta V/dt = i_O$.